Direct finite-time entropy sampling for the three-mode NLS chain

Numerical audit report

1 September 2026

1 Question and frozen calculation

The calculation tests whether ordinary sampling resolves both signs of the finite-time entropy observable for the smallest chain with an unforced middle mode. The parameters are

$$n = 3, \qquad T_L = 10, \quad T_R = 2, \quad \gamma = 0.1, \quad \Delta t = 5 \times 10^{-4}.$$

After burn-in time 500, the simulation records 7,813 non-overlapping blocks of length 20 in each of 128 independent streams, giving 1,000,064 base blocks. The accumulated observables are

$$\Sigma^{\mathrm{m}} = -\frac{Q_L}{T_L} - \frac{Q_R}{T_R}, \qquad R_E = Q_L + Q_R - \Delta E,$$

together with the middle-bond action current and the five-dimensional reduced state at both block endpoints. Longer horizons are formed only by grouping consecutive blocks within the same stream.

The mean rates in the base data are

$$\langle Q_L/t \rangle = 6.22090943, \quad \langle Q_R/t \rangle = -6.22091138, \quad \langle \Sigma^{\mathrm{m}}/t \rangle = 2.48836475,$$

and the mean action current is 1.25228320. All eight batches completed, with zero midpoint failures.

2 Raw negative-tail resolution

t	blocks	$\#\{\Sigma^{\mathrm{m}} < 0\}$	$P(\Sigma^{\mathrm{m}} < 0)$	95% stream-bootstrap CI	paired bins
20	1,000,064	41	4.09974×10^{-5}	$[2.89981, 5.39965] \times 10^{-5}$	0
40	499,968	0	0	$[0, 0]$	0
60	333,312	0	0	$[0, 0]$	0
80	249,984	0	0	$[0, 0]$	0
100	199,936	0	0	$[0, 0]$	0
120	166,656	0	0	$[0, 0]$	0
140	142,848	0	0	$[0, 0]$	0
160	124,928	0	0	$[0, 0]$	0
180	111,104	0	0	$[0, 0]$	0
200	99,968	0	0	$[0, 0]$	0

Table 1: Raw negative-medium-entropy counts. "Paired bins" is the number of predeclared symmetric Freedman–Diaconis bin pairs having at least 20 raw counts on each side.

The predeclared direct-resolution gate requires at least 1,000 negative blocks and at least eight symmetric bin pairs with 20 counts on both sides. No horizon passes. Consequently no detailed-symmetry slope, confidence interval, or R^2 can be estimated from these data without replacing raw tail counts by an extrapolation. No fit window was selected and no cloning run was substituted.

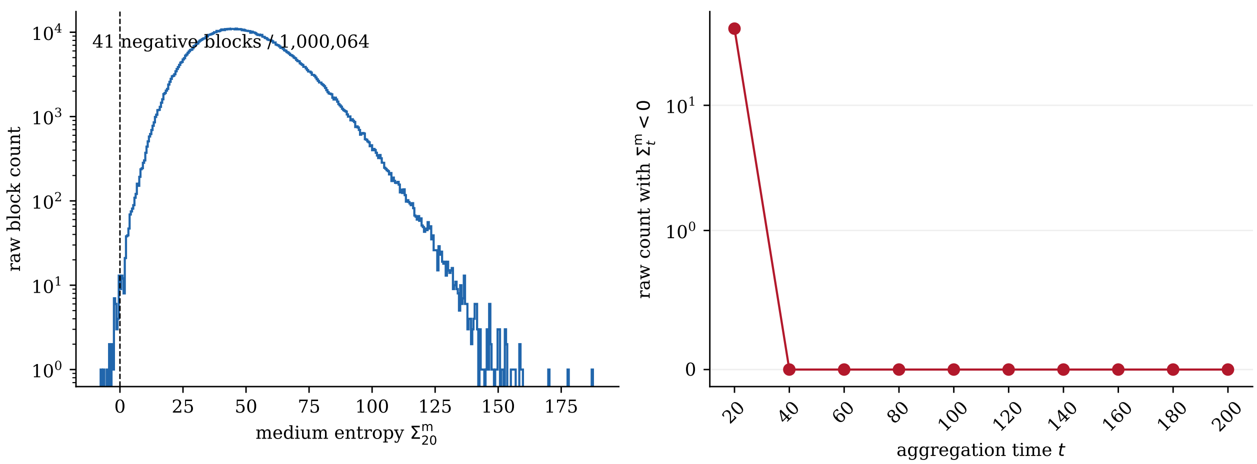


Figure 1: Left: the raw $t = 20$ medium-entropy histogram. Right: negative-event counts after within-stream aggregation.

3 First-law residual

Table 2: Complete reported distribution summary of $R_E = Q_L + Q_R - \Delta E$.

t	mean	std	RMS	min	2.5%	median	97.5%	max
20	-1.417e-5	2.970e-4	2.974e-4	-4.847e-3	-6.146e-4	-1.664e-5	6.077e-4	3.843e-3
40	-2.834e-5	4.187e-4	4.196e-4	-4.997e-3	-8.644e-4	-3.272e-5	8.421e-4	3.994e-3
60	-4.251e-5	5.124e-4	5.142e-4	-5.463e-3	-1.057e-3	-4.814e-5	1.011e-3	4.172e-3
80	-5.668e-5	5.907e-4	5.934e-4	-5.268e-3	-1.229e-3	-6.281e-5	1.153e-3	4.313e-3
100	-7.080e-5	6.612e-4	6.650e-4	-5.750e-3	-1.373e-3	-7.771e-5	1.276e-3	4.642e-3
120	-8.502e-5	7.228e-4	7.278e-4	-5.629e-3	-1.508e-3	-9.181e-5	1.380e-3	4.520e-3
140	-9.919e-5	7.825e-4	7.888e-4	-5.447e-3	-1.641e-3	-1.073e-4	1.481e-3	4.270e-3
160	-1.133e-4	8.352e-4	8.428e-4	-5.493e-3	-1.759e-3	-1.204e-4	1.561e-3	4.755e-3
180	-1.275e-4	8.856e-4	8.948e-4	-5.179e-3	-1.863e-3	-1.367e-4	1.657e-3	4.336e-3
200	-1.416e-4	9.347e-4	9.454e-4	-5.267e-3	-1.972e-3	-1.478e-4	1.736e-3	4.418e-3

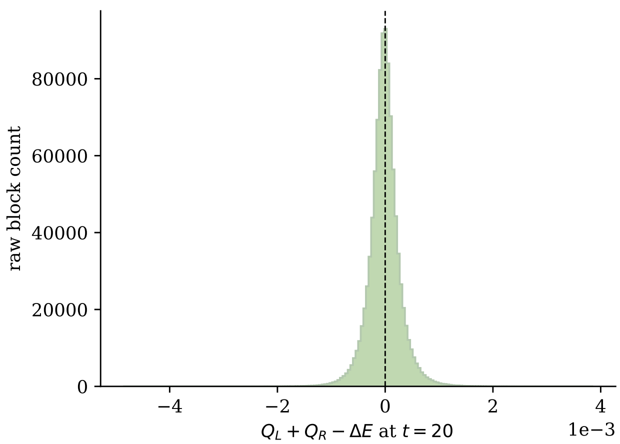
At $t = 20$, the residual RMS is 2.97383×10^{-4}, or 2.24426×10^{-6} of the RMS left-bath heat. Entropy and balance columns recompute from the saved heat and energy fields within binary64/CSV roundoff; all consecutive reduced endpoints agree exactly after parsing.

4 Medium-only integral diagnostic

t	$\log\langle e^{-\Sigma^{\mathrm{m}}}\rangle$	95% CI	weight ESS	largest share
20	-5.7736	$[-8.0682, -4.8381]$	2.288	0.6155
40	-27.7021	$[-29.0827, -27.0673]$	4.967	0.3126
60	-52.4555	$[-54.2741, -51.7400]$	3.780	0.3542
80	-73.5308	$[-87.8362, -72.4322]$	1.000	1.0000
100	-106.9585	$[-119.3689, -105.6540]$	1.169	0.9214
120	-135.1248	$[-154.3239, -134.0262]$	1.000	1.0000
140	-178.9688	$[-198.5417, -177.8702]$	1.000	1.0000
160	-219.2272	$[-229.2468, -218.1317]$	1.030	0.9852
180	-246.6478	$[-265.0109, -245.5492]$	1.000	0.9999
200	-299.5255	$[-307.3934, -298.4274]$	1.188	0.9135

Table 3: Medium-entropy-only exponential averages. ESS and largest weight share show that these averages are controlled by only one to five effective samples and are not resolved rare-event estimators.

These values are reported as requested but are not a total-entropy integral FT test: the system-entropy endpoint contribution is omitted, and the exponential average has extremely low tail ESS.



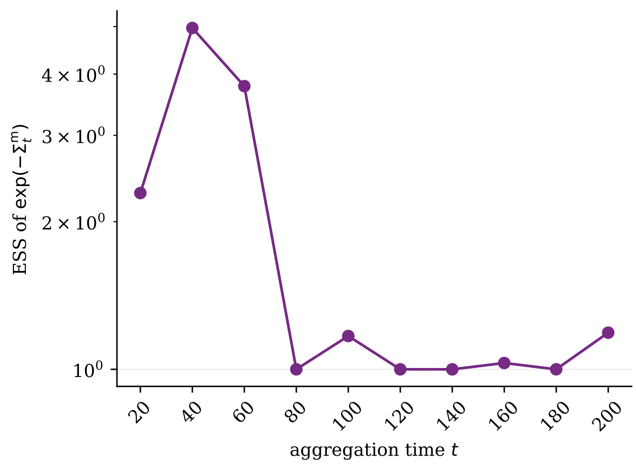


Figure 2: Left: raw $t = 20$ first-law residuals. Right: effective sample size of the medium-entropy exponential weights.

5 Stationary-density limitation and provenance

The saved Section-4 network `KDE/4:15_NN/h5_files/final.keras` has SHA-256 `906f5e60...bc07d536`. Its executed notebook uses $(T_1, T_3) = (2, 8)$, not $(10, 2)$. It also reports scaled log-density RMSE 0.409876 and mean normalized Fokker–Planck residual 0.3733. Applying that network to these endpoints would not provide the requested stationary density. Therefore Δs_{sys}, the total-entropy detailed slope, and the total-entropy integral FT are not evaluated.

The production executable used Git commit `9fd7d75a45a45473a82a380e615f7b97c4a4176c`, source SHA-256 `9ae5835e...6d4eaf2`, binary SHA-256 `15436d39...7be18e`, and seed 2026083133. The raw block file has SHA-256 `4f728b3d...7d7a088f`. The exact command and complete hashes are recorded in `provenance/production_manifest.txt`.

6 Strict outcome

Direct sampling at $n = 3$ does not resolve the negative tail sufficiently for a detailed finite-time FT fit: only 41 negative blocks occur at $t = 20$, and none occur at $t \geq 40$. This is a support limitation, not evidence for or against the fluctuation theorem. Cloning or another rare-event method may now be considered, but none was started as part of this direct-sampling experiment.